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# Intravitreal RTH258 (brolucizumab) demonstrates no effect on pregnancy, parturition, embryofetal or postnatal development in cynomolgus monkeys

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ABSTRACT

RTH258 (brolucizumab) is a humanized single chain antibody fragment, the smallest functional unit of an antibody designed to target vascular endothelial growth factor in angiogenic retinal disease. To further understand the safe use of RTH258, this study assessed the potential impact of intravitreal RTH258 on pre- and postnatal development in the offspring of cynomolgus monkeys following administration to the mother. Three groups of 16 pregnant females were included: a low dose group (RTH258 3 mg/50 µl [60 mg/ml]), a high dose group (RTH258 6 mg/50 µl [120 mg/ml]), and a control group. Maternal animals were administered a single injection of 50 µl in the right eye once every four weeks. Animals were observed daily and detailed observations were collected before and after the first dose, and then weekly thereafter. Following parturition, observations of infants included external, morphological, and ophthalmic examinations; neurobehavioral test battery; grip strength; and skeletal development. Blood samples for hematology, coagulation, and clinical chemistry were collected from non-fasted maternal and infant animals. No RTH258-related deaths occurred in maternal dams or infants. No RTH258-related clinical observations were noted in maternal animals or in surviving infants - there were no changes in gestation length; pregnancy loss; deaths; body weight/weight change; infant grip strength; infant external, morphological, or skeletal evaluations; ophthalmoscopy or neurobehavioral observations; or clinical pathology parameters. RTH258 had no impact on pregnancy or parturition; embryo-fetal development; or survival, growth, or postnatal development of offspring when administered via repeated intravitreal administration.

## 1. Introduction

RTH258 (brolucizumab) belongs to a class of drugs that target vascular endothelial growth factor (VEGF). In several retinal diseases, including neovascular age-related macular degeneration (nAMD) and diabetic macular edema (DME), VEGF is upregulated and leads to uncontrolled growth of highly permeable blood vessels in the choroid, in a process known as neovascularization [1]. Leakage of fluid from these abnormal blood vessels beneath the retinal pigment epithelium, into the sub-retinal space, and within the retina itself has a significant negative impact on retinal structure, physiology, and, consequentially, on visual function [1]. RTH258, and other molecules like it, aim to inhibit the activity of VEGF, preventing the growth of abnormal blood vessels, and leading to fluid resolution and restoration/maintenance of vision [2–4].

RTH258 was first approved for use in nAMD, a population that is exclusively beyond reproductive age. Before such a VEGF inhibitor could be approved for use in younger (potentially reproductively-active) populations, e.g., diabetic retinopathy patients, the relevant health authorities required additional data on the pre-natal and postnatal developmental toxicity effect of the drug.

During the process of fetal development, most proteins are too large to diffuse across the placenta [5]. However, placental transfer of antibodies is a process that provides an evolutionary advantage and as such, in the primate, mechanisms have evolved to enable this. The transfer of antibodies across the placenta allows acquisition of passive immunity in the fetus, enabling the infant to have a degree of humoral immunity at birth, and is central to the provision of immunity in early life [6].

Placental transfer of antibodies is limited to IgG serotypes – and even

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within the IgG antibody class, there is variability in degree of transfer dependent on subclass (IgG1>IgG4>IgG3>IgG2) [7]. Transfer of antibodies across the placenta is mediated by the FcRn receptor - the same receptor that is responsible for the regulation of antibody serum persistence [8]. For placental transfer to occur, the Fc region of an IgG or Fc fusion protein binds to the FcRn receptor on the syncytiotrophoblast cells of the chorioallantois of the primate placenta, leading to receptor-mediated transcytosis and transfer to the fetal circulation. Antibody fragments that lack the Fc domain are not transported across the placenta, limiting embryo-fetal exposure, and lowering the risk of direct embryo-fetal toxicity [9–11]. Indirect effects of macromolecules that do not contain an Fc domain may still occur secondary to altered maternal health, homeostasis, or placental function. In addition, rarely, if anti-drug antibodies (ADA) develop and bind to the antibody fragments, the Fc region of the bound ADA may enable placental transfer to occur [12].

RTH258 is a humanized single-chain antibody fragment (scFv) with the lowest molecular weight of VEGF inhibitors [13]. The smaller size of this scFv (26 kDa) is notable in comparison with other anti-VEGF therapies such as bevacizumab (a full length IgG1 antibody, approximately 150 kDa), ranibizumab (a Fab fragment, approximately 50 kDa), aflibercept (a VEGFR1/2-Fc fusion protein, 97–115 kDa) and faricimab (a bispecific antibody, approximately 150 kDa) (Fig. 1) [14]. When assessing the reproductive toxicity of biotherapeutics, the majority of placental transfer of antibodies in primates occurs in the third trimester of pregnancy, with only minimal transfer occurring during organogenesis [15,16]. Consequently, in the absence of overt maternal toxicity, the greatest period of susceptibility to antibody-based therapeutics is the fetal period (growth and maturation) rather than embryogenesis [17].

For molecules that are systemically-administered, an Fc domain is generally advantageous, enabling exploitation of FcRn-mediated recycling systems, which prolong systemic exposure. However, for a molecule administered intravitreally, with an intraocular target, systemic exposure is not desired, and the low molecular weight and absence of an Fc domain allow more rapid systemic clearance.

Studies with RTH258 in non-human primates have shown that its low molecular weight leads to fast systemic clearance potentially reducing the risk of systemic side effects [13,14,18]. In addition to rapid systemic clearance in patients, in Phase III trials for the treatment of nAMD and DME, RTH258 demonstrated clinically meaningful visual acuity gains and excellent anatomic improvements [2,3,19].

New medicinal products must be assessed for developmental and reproductive toxicity (DART) in accordance with harmonized tripartite guidelines that provide recommendations on the design and timing of DART studies [20]. The rabbit and rodent, the most commonly utilized species' for the assessment of reproductive toxicity, were unsuitable for RTH258, as the intravitreal route of administration was not feasible in the rodent (due to the small vitreous volume and short vitreous half-life) [21], and repeat dose administration of intravitreal biologics in the rabbit are frequently confounded by severe ocular inflammation secondary to the development of systemic anti-drug antibodies (ADA) [22]. ICH S6(R1) specifies an enhanced pre- and postnatal development (ePPND) study design in the non-human primate (NHP), which may be used to evaluate all aspects of developmental toxicity for a biopharmaceutical [23].

This ePPND study with RTH258 was conducted on the basis of a health authority request, triggered by the expansion of approved indications to DME, in which the patient population includes women of childbearing potential.

In this ePPND study, pregnant monkeys were allowed to give birth and raise their infants for 12 weeks post-partum (p.p); with a morphological examination of the neonates at birth [24–26]. Based on previous ePPND studies [27], a minimum of 14 pregnant animals per group was considered sufficient to predictably achieve six to eight live infants per study group on postnatal day (PND) 7 as recommended by ICH S6(R1). Pregnancy outcome in the cynomolgus monkey can be predicted statistically by using group-size-related normograms based on variability of pregnancy success data experienced in over 60 studies in pregnant animals [28].

The purpose of this non-clinical laboratory study was to evaluate the potential effects of intravitreal (IVT) RTH258 on pregnancy and parturition, lactational transfer, embryo-fetal development, survival, growth, and postnatal development of offspring when administered to the mother via repeated dosing (once every four weeks until delivery, or once every four weeks until delivery and on Day 28 p.p.).

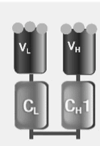
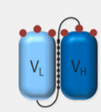
| | Ranibizumab[2,3] | Aflibercept[1] | Brolucizumab[4,5] | Unlicensed bevacizumab[7] | Faricimab[6] |
|---|---|---|---|---|---|
| Format[1-5] | Fab fragment | VEGFR1/2-Fc fusion protein | Single-chain antibody fragment | Full antibody (IgG1) | Bispecific antibody |
| Molecular structure |  |  |  |  |   |
| Molecular weight[1-5] | ~48 kDa | 97-115 kDa[a] | 26 kDa | ~149 kDa | ~150 kDa |
| Clinical dose[2,3,5-7] | 0.50 mg | 2.00 mg | 6.00 mg | 1.25 mg | 6.00 mg |
| Relative number of molecules per injection | **0.5-0.6** | **1.0** | **11.2-13.3** | **0.4-0.5** | **2.0-2.4*** |

**Fig. 1.** Anti-VEGF antibodies currently in clinical use for treating retinal diseases. [a] Molecular weight expressed as a range to reflect glycosylation status. *Faricimab 6 mg is 4 times molar dose of 0.5 mg ranibizumab, kDa, kilodalton; VEGF, vascular endothelial growth factor, 1. Eylea Summary of product characteristics. Available at https://www.ema.europa.eu/en/documents/product-information/eylea-epar-product-information_en.pdf. Accessed on April 07,2021; 2. Lucentis Summary of product characteristics. Available at https://ec.europa.eu/health/documents/community-register/2007/2007012218294/anx_18294_en.pdf. Accessed on April 07,2021; 3. CATT Research Group. N Engl J Med. 2011;364:1897–1908; 4. Holz FG, et al. Ophthalmology. 2016;123:1080–1089; 5. Dugel PU, et al. Ophthalmology 2017;124:1296304; 6. Sahni, J et al. Ophthalmology 2019;126:1155–70; 7. Avastin Summary of product characteristics. Available at https://www.ema.europa.eu/en/documents/product-information/avastin-epar-product-information_en.pdf. Accessed on April 07,2021.

## 2. Methods

### 2.1. Animal welfare, care, and use statement

The study was conducted in an AAALAC accredited facility. All procedures in the protocol complied with the Animal Welfare Act, the Guide for the Care and Use of Laboratory Animals, the Office of Laboratory Animal Welfare and were approved by the local Institutional Animal Care and Use Committee (IACUC). The protocol was reviewed by the Novartis Primate Ethical Review Committee (NPERC).

Forty-eight female cynomolgus monkeys of Asian origin (drug naïve) were received from Labcorp Research Products, Inc. (Alice, TX, USA). Animals were acclimated to the test facility for at least 4 weeks prior to mating for study assignment. At initiation of dosing, animals were 4–6 years old, and body weights ranged from 2.5 to 6.1 kg. Animals were group-housed in European guideline (ETS 123) compliant pens (up to three maternal animals with their offspring/pen) that were enriched with a bedding material. Water was provided ad libitum and animals were offered certified Primate Diet #5L4L (PMI Nutrition International Certified LabDiet®) at least once daily (a.m.), unless fasted for study procedures. A supplemental diet enriched with Vitamin $D_3$ (PMI #5040) was given once daily (p.m.).

### 2.2. Determination of pregnancy

Sexually mature males were used for mating only. Females were paired with fertile males for a maximum of 48 h at the time of expected ovulation. Ultrasonographic examinations were done on Days 18–20 post-coitum (p.c.) to confirm pregnancy. Females were considered pregnant when the lumen of the uterus was detectable as hypoechoic focus (the first day of mating was designated as Day 0 p.c.).

### 2.3. Test and control articles

RTH258 was provided in two concentrations, RTH258 3 mg/50 µl (60 mg/ml) (Batch 2019600) and RTH258 6 mg/50 µl (120 mg/ml) (Batch 2022748). The control article was RTH258 placebo (0.05 ml [50 µl]), composed of sucrose (Batch No. S0005), citric acid monohydrate (Batch No. S0015/S0016), sodium citrate dihydrate (Batch No. S0017), polysorbate 80 low peroxide (Batch No. S0022/S0023), and sterile water for injection (Batch No. J7S004/J8D069). RTH258 vials were removed from refrigerated storage just before dosing syringes were filled. Dosing syringes were fitted with a sterile, 30-gauge x ½ inch needle for the IVT injection.

### 2.4. Study design

The study design is detailed in Fig. 2. The study included three groups: a low dose group, a high dose group and a control group, each containing 16 female animals. The low dose group received RTH258 3 mg/50 µl (60 mg/ml) and the high dose group received RTH258 6 mg/50 µl (120 mg/ml). The control group received a 50 µl dose of RTH258 placebo. The dose and dose intervals selected support those utilized in humans and were well tolerated in previously conducted IVT toxicity studies in the non-human primate, with no signs of ocular or systemic toxicity [29]. Animals were administered a single injection of 50 µl in the right eye once every four weeks starting on Day 20 p.c. until delivery (Days 20, 48, 76, 104, 132, and 160 p.c.). Animals were anesthetized with intramuscular injections of atropine (0.01 mg/kg), ketamine (2–10 mg/kg), and dexmedetomidine (0.25 mg/kg) prior to dosing. Following the dose procedure, the anesthesia was reversed with atipamezole (0.25 mg/kg) as needed. Four maternal animals/group with infants surviving on Day 28 p.p. received an additional dose of RTH258 or RTH258 placebo on Day 28 p.p. and had blood and milk samples collected for toxicokinetic evaluation (approximately 6 [blood only] and 24 h post dose).

### 2.5. Clinical observations of maternal animals and embryo-fetal growth

Cage-side observations were also conducted for each animal once daily, with any abnormal findings recorded. Detailed observations were collected prior to the first dose on Day 20 p.c., on Day 27 p.c., and weekly thereafter based on Day 27 p.c. Additionally, detailed observations were collected on Days 1 and 7 p.p., weekly thereafter, and on the day of final disposition. Ultrasonography examinations, including assessment of fetal viability, were performed on Days 27, 41, 55, 69, 83, 97, 111, 125, 139, 153, 156, 160, 163, 167, 170, 174, 177, and 181 p.c. or until delivery. When signs of abortion were observed, additional ultrasonography was performed. Additional ultrasounds were collected on Day 1 p.p. for confirmation of delivery (due to group housing). Ultrasound for confirmation of delivery occurred on Day 0 p.p. in urgent cases when infant identification was required. Vaginal swab data were recorded once daily until delivery. Upon each delivery, the placenta and umbilical cord (when available) were collected and. delivered to the Labcorp pathology department for preservation.

### 2.6. Body weight of maternal animals and infants

Body weights of maternal animals were recorded prior to the first

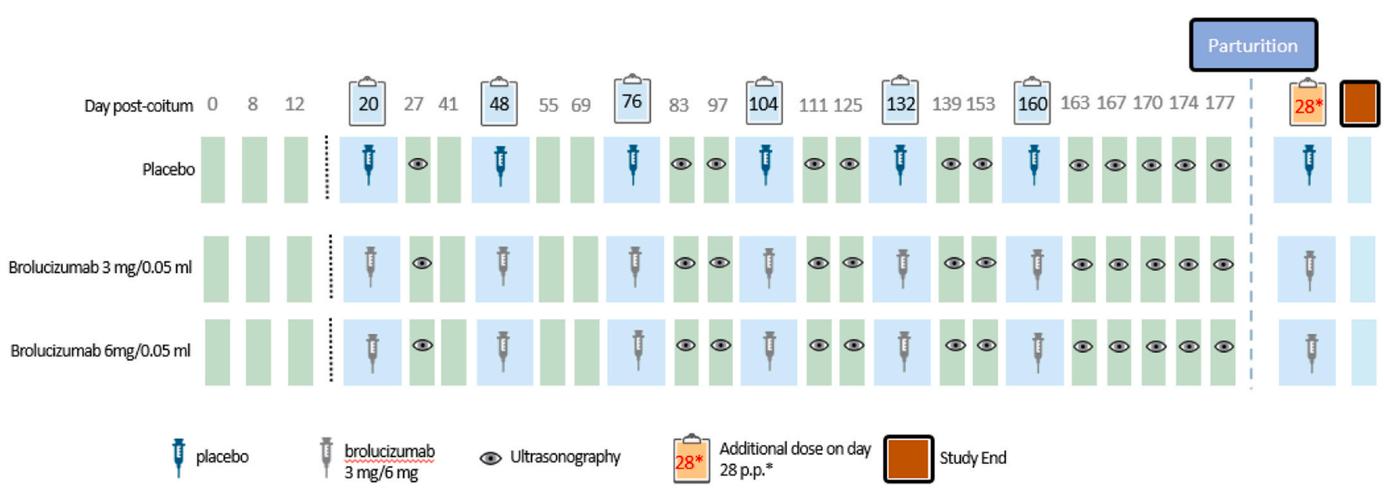



**Fig. 2.** RTH258 ePPND Study Design. ePPND, enhanced pre and postnatal development; p.p. post-partum, Ultrasonography examinations included assessment of vitality. When signs of abortion were observed, additional ultrasonography was performed. * Four maternal animals/group with infants surviving on Day 28 p.p. received an additional dose and had blood and milk samples collected for toxicokinetic evaluation.

dose on Day 20 p.c., on Day 27 p.c., and weekly thereafter based on Day 27 p.c. Additionally, body weights were collected on Days 1 and 7 p.p., weekly thereafter based on Day 7 p.p., and on the day of final disposition. Body weights of infants were recorded on PND 1 and 7, weekly thereafter, and on the day of scheduled necropsy.

### 2.7. Clinical observations of infants

Cage-side observations were conducted for each infant once daily, with abnormal findings recorded. Detailed observations were conducted for each infant once on PND 1 and 7, weekly thereafter, and on the day of scheduled necropsy. Abnormal findings or an indication of normal was recorded. External examinations (PND 1 [including sex determination], 7, 14, 21, and 28 and then every 4 weeks thereafter for up to 3 months), morphological examinations (PND 1, 21, 56, and 84), neurobehavioral test battery (PND 1 and 7), and grip strength (PND 28) were also performed. Skeletal development was assessed via X-ray on PND 50 $\pm$ 2 days. Infants were anesthetized with ketamine/dexmedetomidine prior to a single X-ray image being taken for each infant. The infants were administered an anesthetic reversal agent (atipamezole) following image collections.

### 2.8. Ophthalmological examinations of maternal animals

Ophthalmic examinations were conducted on anesthetized maternal animals once prior to the first dose and on days of dosing (prior to dosing). Pupils were dilated with a mydriatic agent (1% tropicamide) prior to examination by a Board-Certified Veterinary Ophthalmologist using an indirect ophthalmoscope and slit lamp biomicroscope. The adnexa and anterior portion of both eyes were examined using a slit-lamp biomicroscope. The ocular fundus of both eyes was examined (where visible) using an indirect ophthalmoscope.

### 2.9. Clinical laboratory assessments

#### 2.9.1. Sample collection and handling

Blood samples for hematology, coagulation, and clinical chemistry were collected from nonfasted maternal and infant animals via a femoral or cephalic vein. Blood samples were collected from maternal animals on Days 20 and 132 p.c. and Day 91 p.p. and from infants on PND 42 (hematology – approximately 350 μl), PND 56 (clinical chemistry - approximately 1.0 ml), and PND 70 (coagulation - approximately 0.5 ml).

#### 2.9.2. Hematology

Blood samples for hematology analysis were stored no longer than 24 h at room temperature and $\leq$ 72 h at 2–8 °C. Blood samples were checked for clotting prior to analysis. Blood smears were prepared on the same day as blood collection. Smears from high-dose and control animals were reviewed for any change in morphology of red or white blood cells and platelets. The tests conducted were: red blood cell (erythrocyte) count; hemoglobin; hematocrit; mean corpuscular volume; mean corpuscular hemoglobin; mean corpuscular hemoglobin concentration; reticulocyte count (absolute count and relative percentage); blood cell morphology; platelet count; white blood cell (leukocyte) count; absolute neutrophil count; absolute lymphocyte count; absolute monocyte count; absolute eosinophil count; absolute basophil count; absolute large unstained cell count; and blood smear.

#### 2.9.3. Coagulation

Plasma for coagulation analysis was separated within 75 min of blood collection. Samples were analyzed on the same day as blood was collected and were stored at room temperature for up to 8 h. The tests conducted were: prothrombin time; fibrinogen; and activated partial thromboplastin time.

#### 2.9.4. Clinical chemistry

Serum for clinical chemistry analysis was separated from cells within 90 min of blood collection with the use of gel tubes. Samples were analyzed on the same day as blood was collected and were stored at room temperature. Serum levels of the following were analyzed: glucose; urea nitrogen; creatinine; total protein; albumin; globulin; albumin:globulin ratio; total cholesterol; triglycerides; total bilirubin; aspartate aminotransferase; alanine aminotransferase; alkaline phosphatase; creatine kinase; calcium; inorganic phosphorus; sodium; potassium; chloride; and magnesium.

### 2.10. Terminal procedures – maternal

#### 2.10.1. Maternal terminal procedures

On Day 92 p.p. $\pm$ 1 day, all surviving animals, having been fasted overnight, were anesthetized with sodium pentobarbital, exsanguinated, and necropsied. A macroscopic examination of the external features of the carcass; external body orifices; abdominal, thoracic, and cranial cavities; organs; and tissues was performed. The uterus was opened and examined. When an abortion was noted or an infant died, the maternal animal may have been maintained in the social phase as a social partner for surviving animals housed in the same pen. All surviving animals in the social phase, having been fasted overnight, were anesthetized with sodium pentobarbital, euthanized, and necropsied on the day the last maternal animal in the pen was necropsied or at the discretion of the study director.

#### 2.10.2. Infant terminal procedures

2.10.2 Infant terminal procedures On PND 92 $\pm$ 1 day, all surviving infants, having been fasted overnight (with access to maternal milk), were anesthetized with sodium pentobarbital via intravenous injection, exsanguinated, and necropsied on the same day as the maternal animal. Terminal body weights were recorded for euthanized animals. A macroscopic examination of the external features of the carcass; external body orifices; abdominal, thoracic, and cranial cavities; organs; and tissues was performed. Organ weights (as appropriate) were collected, and tissues retained.

### 2.11. Anti-drug antibody analysis

For the purpose of ADA analysis, maternal animals (non-fasted) had blood samples (approximately 1.0 ml) collected via a femoral vein on Days 20 and 132 p.c., prior to dosing. Additionally, maternal animals (non-fasted) had blood samples (approximately 1.0 ml) collected via a femoral vein on Day 28 p.p. and once on Day 91 p.p. Maternal animals that were not administered a dose on Day 28 p.p. had samples collected once. Maternal animals administered a dose on Day 28 p.p. had samples collected prior to dosing and approximately 24 h post-dose. Infants (non-fasted) had blood samples (approximately 1.0 ml) collected via a femoral or cephalic vein once on PND 91. Serum samples were analyzed for antibodies to RTH258 by Labcorp Bioanalytical Services, LLC (Indianapolis, Indiana, United States), using a validated method (Method ECL-0303). The analysis of samples was carried out according to test site standard operating procedures and the rule of applicable regulations.

### 2.12. Bioanalytical and toxicokinetic analysis

Blood samples for bioanalysis were collected from all maternal animals, when possible, on Days 48 p.c. and 132 p.c. pre-dose and approximately 6 and 24 h post-dose and were collected from 4 maternal animals/group on Day 28 p.p approximately 6 and 24 h post-dose. Blood samples were also collected from all remaining non-dose maternal animals on Day 28 p.p. and from all maternal animals on Day 91 p.p. Blood samples were also collected from two infants/sex/group on PND 28 approximately 24 h after the maternal dose. Milk samples were collected from 4 maternal animals/group on Day 28 p.p. approximately 24 h post-dose. Concentrations of free RTH258 in serum were measured using a validated ligand binding assay (Method No. ECL-0302). Also, blood was collected from all animals on Day 20 p.c. and analyzed for monkey chorionic gonadotropin (mCG) from 5 animals that were presumed pregnant. Toxicokinetic parameters were estimated using a non-compartmental approach consistent with the extravascular route of administration. The individual serum concentration-time data was used for toxicokinetic calculations. In addition to parameter estimates for individual animals, descriptive statistics (e.g. mean and standard deviation) was reported, as appropriate. All toxicokinetic parameters were generated from individual RTH258 concentrations in serum. Toxicokinetic analysis included parameters listed in Supplementary Table 1.

### 2.13. Data evaluation and statistical analysis

Only data collected on or after the first day of dosing were analyzed statistically. Analysis of variance (ANOVA) and pairwise comparisons were used to analyze the following: absolute body weight (maternal and infant); body weight change (maternal and infant); morphological data (infant); continuous clinical pathology data (maternal and infant); terminal body weight (infant); absolute organ weight; organ:body weight; and organ:brain weight (infant).

Levene's test was done to test for equality of variances between groups. Where Levene's test was significant ($P \leq 0.05$), a rank transformation (to stabilize the variances) was applied before ANOVA was conducted (note: Levene's test was not applied to the rank-transformed data). Where Levene's test was not significant ($P > 0.05$), ANOVA was conducted.

One-way ANOVA was used (if applicable) to analyze continuous clinical pathology values, organ weight data, body weight data, and infant morphology data. If the group effect of the ANOVA was significant ($P \leq 0.05$), Dunnett's t-test was used for pairwise comparisons between each treated and control group. Group comparisons were evaluated at the 5.0%, two-tailed probability level. If the ANOVA was not significant ($P > 0.05$), no further analyses were conducted. Reproductive indices, including gestation, live birth, viability, and weaning indices and sex ratio (the proportion of male infants) were analyzed using Fisher's exact test (one side decreasing).

## 3. Results

### 3.1. Animal fate

No RTH258-related deaths occurred. One maternal animal administered 3 mg/eye (low dose group) was euthanized in a moribund condition on gestation day (GD) 133. No abnormal observations were noted for the fetus. The edema that led to the moribund euthanasia of this animal was likely due to pregnancy complications and was not considered RTH258-related.

A summary of pregnancy outcomes is shown in Table 1. Because of the high and variable rate of pregnancy loss in cynomolgus macaques, normograms mentioned in the ICH S6(R1) guideline were used to permit a statistically based evaluation of the incidence of embryo-fetal and infant losses (concurrent and historic) in line with industry standards [28]. The total number of surviving infants on PND 7 for each group included 11 from maternal control animals (of 13 infants delivered), 14 from maternal animals administered 3 mg/eye (of 14 infants delivered), and 8 from maternal animals administered 6 mg/eye (of 9 infants delivered) (Table 1). In the control group, in the first two intervals (Days 21–50 p.c.) two abortions occurred. No stillbirths occurred in the control group. No pregnancy losses, abortions or stillbirths occurred in maternal animals administered 3 mg/eye – which is a lower incidence of loss than usually observed in cynomolgus monkeys. In maternal animals administered 6 mg/eye, two abortions occurred in the 3rd and 4th intervals (Days 51–100 p.c.), and subsequently two stillbirths occurred.

**Table 1**
Summary of pregnancy outcomes: Incidence of abortions, stillbirths, and early infant deaths.

| | Control group No. (%), unless otherwise stated | Low dose group 3 mg/eye No. (%) | High dose group 6 mg/eye No. (%) |
|---|---|---|---|
| Total number of pregnant females[a] | 15[a] | 14[a,b] | 13[a] |
| Mean length of gestation (days) | 162 | 165 | 166 |
| Females with abortion | | | |
| 1st Trimester Abortions (GD 0–50) | 2 (13.3) GD 27, 41 | 0 | 0 |
| 2nd Trimester Abortions (GD 51–100) | 0 | 0 | 2 (15.4) GD 65, 83 |
| 3rd Trimester Abortions (GD 101–144) | 0 | 0 | 0 |
| Total number of stillbirths (GD 150+) | 0 | 0 | 2 (15.4) GD 166, 177 |
| Total number of abortions and stillbirths | 2 (13.3) | 0 | 4 (30.8) |
| Total number delivered | 13 (86.7) | 14 (100) | 9 (69.2) |
| Total number of early infant deaths (PND 0–7) | 2 (13.3) PND 0, 1 | 0[b] | 1 (7.7) PND 0 |
| Total number of infants euthanized or found dead after PND 7 | 1 (6.7)[c] PND 21 | 2 (14.3)[d,e] PND 10, 16 | 1 (7.7)[f] PND 18 |
| Total number of surviving infants to terminal sacrifice | 10 (66.7) | 12 (85.7) | 7 (53.8) |

GD, gestation day; PND, postnatal day
[a] Maternal animals confirmed not pregnant by monkey chorionic gonadotropin (mCG) analysis were not included in the total number
[b] Maternal animal P0113 euthanized in moribund condition on GD 133, and fetus, not included in the total number
[c] Infant P0015-1 found dead on PND 21
[d] Infant P0104-1 euthanized on PND 16 due to injury
[e] Infant P0110-1 euthanized on PND 10 due to failure to thrive
[f] Infant P0207-1 euthanized on PND 18 due to failure to thrive likely due to poor maternal milk production.

Incidences of abortions and stillbirths in control and 6 mg/eye group were within the normal range, as indicated on the normogram (Fig. 3a and b) and were not attributed to RTH258.

Infant mortality, not attributed to maternal exposure to RTH258, was observed for infants from all groups. Of the two infants from maternal animals administered 6 mg/eye that were stillborn, one of these infants was partially cannibalized and recorded as having 'bilateral absent metatarsals' and 'absent distal tail'. Three infants from maternal control animals (two during PND 0–7, 1 euthanized or found dead after PND 7), two infants from a maternal animal administered 3 mg/eye (both euthanized or found dead after PND 7), and two infants from a maternal animal administered 6 mg/eye (one during PND 0–7 and one euthanized or found dead after PND 7). However, these deaths were not considered RTH258-related as there was no dose relationship and they were within the normal expected range of infant loss, as indicated on the normogram (Fig. 3b).

One mated control female, one mated female administered 3 mg/eye, and three mated females administered 6 mg/eye were identified with false pregnancies. A false pregnancy occurred when a female appeared pregnant during the GD 20 ultrasound but was determined as not pregnant according to a negative ultrasound on GD 27 and low mCG values in serum collected on GD 20. These animals were moved to the social phase after the negative ultrasound on GD 27 and were removed from the total number of pregnant females.

### 3.2. Animal observations

No RTH258-related clinical observations were noted p.c. or p.p. for


**Fig. 3.** a and b: Normograms showing cynomolgus monkey pregnancy outcomes in groups 1 and 3. Explanation for the normal distribution of outcomes: “likely” contains more than 70%, “can occur” contains approximately 20%, “unusual” contains approximately 8% and "unlikely" contains approximately 2% of outcomes. Jarvis P, et al. (2010), Weinbauer GF, et al. (2011a) and Weinbauer GF, et al. (2011b), p.c., post coitum; p.p., post partum.

maternal animals or for surviving infants i.e., no changes in gestation length (the average gestation length was 162, 165, or 166 days for maternal control animals and maternal animals administered 3 or 6 mg/eye, respectively); pregnancy loss (abortions or stillbirths); deaths; maternal or infant clinical observations; body weight; body weight change; infant grip strength; infant external, morphological, skeletal evaluations; ophthalmoscopy or neurobehavioral observations; clinical pathology parameters; organ weights; or macroscopic observations. The most common findings for infants included skin and pelage observations that are common for infant monkeys. No RTH258-related veterinary treatments were needed.

### 3.3. Ophthalmic examinations

No RTH258-related ophthalmic observations were noted for maternal animals. Ophthalmic observations of vitreous cells were noted in all groups, including controls, and were considered to be due to the dosing procedure and not RTH258-related. No ophthalmic observations prevented dosing for any maternal animal. One maternal control animal and two maternal animals administered 3 mg/eye had cortical or nuclear cataracts noted prior to the first dose. These pregnant animals were considered acceptable to use in the study because the cataracts did not impede the view of the fundus or other structures, were unlikely to progress or change in character, and did not interfere with the dosing procedure.

### 3.4. Clinical laboratory assessments

No RTH258-related hematology, coagulation nor clinical chemistry effects were observed in maternal animals or infants.

### 3.5. Terminal evaluations

No RTH258-related organ weight differences nor macroscopic observations were present in maternal animals or infants.

### 3.6. Toxicokinetic parameters

Table 2 shows a summary of mean RTH258 toxicokinetic parameters in maternal monkey serum following once monthly IVT administration during gestation and on Day 28 p.p. All serum and milk concentration values of RTH258 in the control group were below the lower limits of quantitation (< 5.00 ng/ml for serum and < 10.0 ng/ml for milk). Exposure to RTH258 in maternal milk or infant serum was not observed after IVT administration of 3 or 6 mg/eye to maternal animals. Exposure to RTH258 in maternal animals increased with the increase in dose level

**Table 2**
Summary of mean RTH258$C_{max}$, $C_{max}$/D, $AUC_{0-24}$, and $AUC_{0-24}$/D in maternal monkey serum following once monthly intravitreal administration during gestation through Day 28 p.p.

| Dose Group | Dose Level (mg/eye) | Interval | $C_{max}$ (ng/ml) | $C_{max}$/D (ng/ml)/(mg/eye) | $AUC_{0-24}$ (ng·h/ml) | $AUC_{0-24}$/D (ng·h/ml)/(mg/eye) |
|---|---|---|---|---|---|---|
| Low dose | 3 | Day 48 p.c. | 224 | 74.8 | 3420 | 1140 |
| | | Day 132 p.c. | 253 | 84.3 | 4310 | 1440 |
| | | Day 28 p.p. | 248 | 82.7 | 4820 | 1610 |
| High dose | 6 | Day 48 p.c. | 389 | 64.9 | 6090 | 1010 |
| | | Day 132 p.c. | 301 | 50.2 | 5120 | 854 |
| | | Day 28 p.p. | 483 | 80.4 | 8100 | 1350 |

Cmax, maximum observed concentration; Cmax/D, dose normalized maximum concentration, calculated as Cmax/dose; AUC0–24/D, dose normalized AUC0–24, calculated as AUC0–24/dose; p.c., post-coitum; p.p. post-partum
Notes: Animals were administered a single injection of 50 µl in the right eye once every four weeks starting on Day 20 p.c. until delivery (Days 20, 48, 76, 104, 132, and 160 p.c.). Four maternal animals/group with infants surviving on Day 28 p.p received an additional dose on Day 28 p.p.

from 3 to 6 mg/eye on Day 48 p.c., Day 132 p.c., and Day 28 p.p. The increases in RTH258 mean $C_{max}$ and $AUC_{0-24}$ values were approximately dose proportional. Exposure was generally similar after multiple monthly doses when compared with the second dose on Day 48 p.c.

### 3.7. Anti-drug antibodies

In general, maternal animals screened positive for the presence of ADAs: 8/16 in the control group; 14/16 in the 3 mg/eye group (low dose group); and 9/16 in the 6 mg/eye group (high dose group) animals. Infant animals did not screen positive for the presence of ADAs in the control group, but 5/12 infants in the 3 mg/eye group (low dose group) and 1/7 infants in the 6 mg/eye group (high dose group) screened positive for the presence of ADAs. However, no maternal animals showed reduced exposure consistent with an ADA-response, and exposure to RTH258 was not demonstrated in any infant animal.

## 4. Discussion

In this ePPND study, RTH258 had no impact on pregnancy or parturition; embryo-fetal development; or survival, growth, or postnatal development of offspring when administered via repeated IVT administration once every 4 weeks until delivery, or once every 4 weeks until delivery and on Day 28 p.p. (Lactation Day 28) to one eye of pregnant cynomolgus monkeys. At 26kDA, RTH258 exhibits rapid systemic clearance [13,14], is too large to cross the placenta by passive diffusion (a process limited to molecules less than 0.5kDA), and as an antibody fragment lacking an Fc domain, is also unable to utilize FcRn-mediated transcytosis [6]. Consistent with this, exposure of infants to RTH258 was not observed after IVT administration of 3 or 6 mg/eye to maternal animals. In the absence of exposure of the infants, any findings in these animals are considered highly unlikely to be directly attributable to RTH258.

Our data suggest that RTH258 had no impact on pregnancy or parturition; embryo-fetal development; or survival, growth, or postnatal development of offspring. Also, no noticeable difference in gestation length was noted, and for all animals the gestation length was within the normal expected range of 134–184 days [24,30]. No RTH258-related observations were made for infant grip strength, infant external, morphological, or neurobehavioral observations, infant skeletal observations, or infant organ weights. As noted in the Results section, one stillborn infant was partially cannibalized and recorded as having ‘bilateral absent metatarsals’ and ‘absent distal tail’, but this was not related to the test article.

ICH S6(R1) (2011) [23] and ICH S5 (R2) (1994) [31] provide many study design recommendations resulting in considerably more standardized ePPND study designs [26,32], including the recommendation to plan studies such that six to eight live infants are available on PND 7. For this study, the number of surviving infants on PND 7 for each group included 11 from maternal control animals, 14 from maternal animals administered 3 mg/eye, and 8 from maternal animals administered 6 mg/eye, and incidences of abortions and stillbirths were within the normal range or (in the case of Group 2) fewer than would be anticipated.

Exposure to RTH258 in maternal milk or infant serum was not observed after IVT administration of 3 or 6 mg/eye to maternal animals. Serum exposure to RTH258 in maternal animals increased with the increase in dose level from 3 to 6 mg/eye on Day 48 p.c., Day 132 p.c., and Day 28 p.p. The increases in RTH258 mean $C_{max}$ and $AUC_{0-24}$ values were approximately dose proportional. Exposure was generally similar after multiple monthly doses when compared with the second dose on Day 48 p.c. The 6 mg/eye dose level corresponded to mean maximum observed concentration ($C_{max}$) and area under the curve ($AUC_{0-24}$) values of 301 ng/ml and 5120 ng·h/ml, respectively, in maternal animals on Day 132 p.c.The systemic maximal concentration of brolucizumab observed here following IVT administration was approximately 1000-fold less than the trough concentration of a therapeutic dose of intravenously administered anti-VEGF [29]. The low systemic brolucizumab exposure due to rapid clearance, and resulting lack of accumulation, would not be expected to result in pharmacodynamically relevant systemic VEGF inhibition.

In conclusion, RTH258 had no impact on pregnancy or parturition; embryo-fetal development; or survival, growth, or postnatal development of offspring when administered via repeated IVT administration once every 4 weeks until delivery, or once every 4 weeks until delivery and on Day 28 p.p. (Lactation Day 28) to one eye of pregnant cynomolgus monkeys. These findings support the label extension of RTH258 to indications in populations of reproductive potential.

## Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Helen Booler reports a relationship with Novartis Pharma AG that includes: employment. Anthony DeLise reports a relationship with Novartis Pharmaceuticals Corporation that includes: employment. Erik Nimz reports a relationship with Novartis Pharmaceuticals Corporation that includes: employment. Diana Shefchek and Marc Leutjens are employees of Labcorp Early Development Laboratories Inc., US and Labcorp Early Development Services GmbH, Germany, respectively. Labcorp provided analytic services during the study - outlined in the Methods.

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### Disclosures

Helen Booler: Novartis employee; Anthony M. DeLise: Novartis employee; Erik Nimz: Novartis employee; Diana Shefchek: Labcorp employee; C. Marc Luetjens: Labcorp employee.

## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.reprotox.2023.108468.

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